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Expert Answer

General Guidance

Resolving forces:

When the force is inclined, it is then resolved into two components: horizontal component of force and vertical component of force.

When the inclination of force with horizontal axis is known, the horizontal component is the product of force's magnitude and cosine of the inclination and the vertical component is the product of force's magnitude and sine of the inclination.

Similarly, when the inclination of force with vertical axis is known, the horizontal component is the product of force's magnitude and sine of the inclination and the vertical component is the product of force's magnitude and cosine of the inclination.

Frictional force:

It is the force generated by the two surfaces which are in contact and slide against each other.

Static friction:

When two objects are in contact with each other and force or load is applied on anyone of object, then there will be an opposing force which will resist the initial motion of the body. The resisting force for static object which depends upon the coefficient of static friction between the two objects is known as static friction.

Kinetic friction:

When two objects are moving relative to each other .then there will be an opposing force which will resist the motion of the body. The resisting force for moving object which depends upon the coefficient of kinetic friction between the two objects is known as kinetic friction.

Coefficient of static friction:

It is referred as the ratio of the frictional force acting at the body to its normal force. It is denoted by

$$\mu_s$$

and it is dimensionless variable.

Consider the inclined force

$$F$$

acting at the end

$$B$$

of the rod AB as shown in Figure (1).



The horizontal component

$$(F_x)$$

of force is expressed as follows:

$$F_x = F \cos \theta$$

Here, magnitude of the force is

$$F$$

and the angle made by the force with the horizontal axis is

$$\theta$$

.

The vertical component

$$(F_y)$$

of force is expressed as follows:

$$F_y = F \sin \theta$$

Write the equilibrium condition for forces along the

$$x$$

-axis.

$$\sum F_x = 0$$

Here, the sum of forces along x-axis is

$$\sum F_x$$

.

Write the equilibrium condition for forces along the

$$y$$

-axis.

$$\sum F_y = 0$$

Here, the sum of forces along y-axis is

$$\sum F_y$$

.

The formula to calculate the static frictional force is as follows:

$$F = \mu_s N$$

..... (1)

Here, static frictional force is

$$F$$

, coefficient of static friction is

$$\mu_s$$

and normal force is

$$N$$

.

Step: 1

Draw the free body diagram of the crate as in Figure (2).

$$\sum F_y = 0$$

$$B_y - 12 + C_y = 0$$

Here, mass of the crate is

$$m$$

, acceleration due to gravity is

$$g$$

, force is

$$P$$

, friction force is

$$F$$

and normal force is

$$N$$

.

Calculate the angle of inclination

$$(\theta)$$

of force P.

$$\theta = \tan^{-1} \left(\frac{3}{4} \right)$$

$$\theta = 36.86^\circ$$

Apply equilibrium condition for forces along the y-axis.

$$\sum F_y = 0$$

$$N - mg - P(\sin \theta) = 0$$

Substitute

$$50 \text{ kg}$$

for

$$m$$

,

$$9.81 \text{ m/s}^2$$

for

$$g$$

,

$$36.86^\circ$$

for

$$\theta$$

and

$$200 \text{ N}$$

for

$$P$$

.

$$N - (50 \text{ kg}) (9.81 \text{ m/s}^2) - (200 \text{ N}) (\sin 36.86^\circ) = 0$$

$$N = 610.5 \text{ N}$$

Apply equilibrium condition for forces along x-axis.

$$\sum F_x = 0$$

$$F - P (\cos \theta) = 0$$

Substitute

$$200 \text{ N}$$

for

$$P$$

and

$$36.86^\circ$$

for

$$\theta$$

.

$$F - (200 \text{ N}) (\cos 36.86^\circ) = 0$$

$$F = 160 \text{ N}$$

Explanation:

The normal force is determined by applying equilibrium condition for forces along y-axis. The friction force is determined by applying equilibrium condition for forces along x-axis.

Hint:

Use Equation (1) to determine the maximum friction force.

Step: 2

Calculate the maximum friction force.

$$F_{\max} = \mu_s N$$

Substitute

$$0.3$$

for

$$\mu_s$$

and

$$610.5 \text{ N}$$

for

$$N$$

.

$$F_{\max} = 0.3 (610.5\text{N})$$

$$F_{\max} = 183.15\text{N}$$

The friction force is less than the maximum friction force.

$$F < F_{\max}$$

$$160\text{N} < 183.15\text{N}$$

Therefore, the friction force calculated using equilibrium condition of forces is satisfactory.

Explanation:

The numerical values of coefficient of static friction and normal force are substituted in Equation (1). The equation is then simplified to determine the maximum static friction force. As

$$F < F_{\max}$$

, the body will remain in rest since the frictional force acting between the crate and ground is less than the maximum static force.

Answer:

The friction force

$$(F)$$

developed between the crate and ground is

$$160 \text{ N}$$

.